National University of Computer and Emerging Sciences, Lahore Campus

Course:	**Software Testing**	**Course Code:**	**CS497**
Program:	**BS (CS)**	**Semester:**	**Spring 2020**
Duration:	**240 Minutes (4 Hours)**	**Total Marks:**	**60**
Paper Date:	**24-Jun-20**	**Weight**	**40%**
Section:	**All**	**Page(s):**	**6**
Exam:	**Final**	**Roll No:**	

Instruction/Notes:

1. This is an open books and open notes exam which should be attempted in student's own handwriting. Attempt all questions in context of class discussions, handouts, text book(s) etc.
2. State your assumptions clearly. Start every question on a new page.
3. Submit through slate.There will be 3 resubmissions allowed. No more resubmissions (either through slate or any other means will be considered for grading).
4. Submit single file which should either be a word file or a pdf file. Submission of separate images or a single compressed file is discouraged
5. There are 5 marks for submitting a single word or pdf file. These 5 marks will not be awarded if your final exam submission is not a single word or pdf file.
6. The submission should be in legible handwriting. The complete (submitted) exam should be clear and readable.
7. If there are multiple submissions from a student, then the latest submission within the original deadline will be considered for grading
8. If you have to acknowledge some book, text, or person, provide the acknowledgement on the last page of your exam.
9. Clearly label question numbers and part numbers while providing the answers
10. Write your complete roll number on your exam.
11. Name of the submitted file should have the format STFinalExam _CompleteRollNumber. Your complete roll number in the file name should be like 12L1010
12. In the exam, whenever you are asked to provide design of test cases use a proper test case structure. For example use the following test case structure if there are 3 input variables:

Test Case Id	Test Case Purpose	Input data			Expected Output
		Variable 1	V2	V3	
1	To test invalid value of Variable 1	23	10	10	Show error message

If required in the exam and you think it is appropriate, you should add more columns s.a. Actual Output, Result etc.

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Question 1 **20 Marks**

Bibliography was maintained through manual indexing but over the years it has been realized that manual searching was not efficient and it had to be automated. Software house XYZ has been given the task to automate this system. Firstly requirements have been analyzed for four major functions:

1. Add a Reference 2. Remove a Reference 3. Edit a Reference 4. Search a Reference

The initial stage of the development of the product is analysis of the requirements. Following requirements are extracted in the first run.

1. A reference may be added into the bibliography providing that the reference does not already exist.
2. A reference may be removed from bibliography providing that the reference already exists.
3. A reference may be edited and any information pertaining to that reference may be changed.
4. A reference may be searched and all references matching the criteria are returned.Criteria for searching is given below:
    a) One or more author (last) names.
    b) One or more words present in a title.

Once the above requirements are finalized the next phase is to develop the design of the system.After the design process architecture of the bibliography system is finalized as shown in Figure 1. After the completion of the development process, testing is still pending.

Your task is to suggest (with reason(s)) two "**black box testing"** techniques that are applicable in this scenario. Apply any one of these techniques to develop minimum number of test cases. Provide reason(s) for selecting the applied technique. The test cases should have proper structure and include input values for all variables. The test cases should also clearly express the expected output. The test cases should be complete.

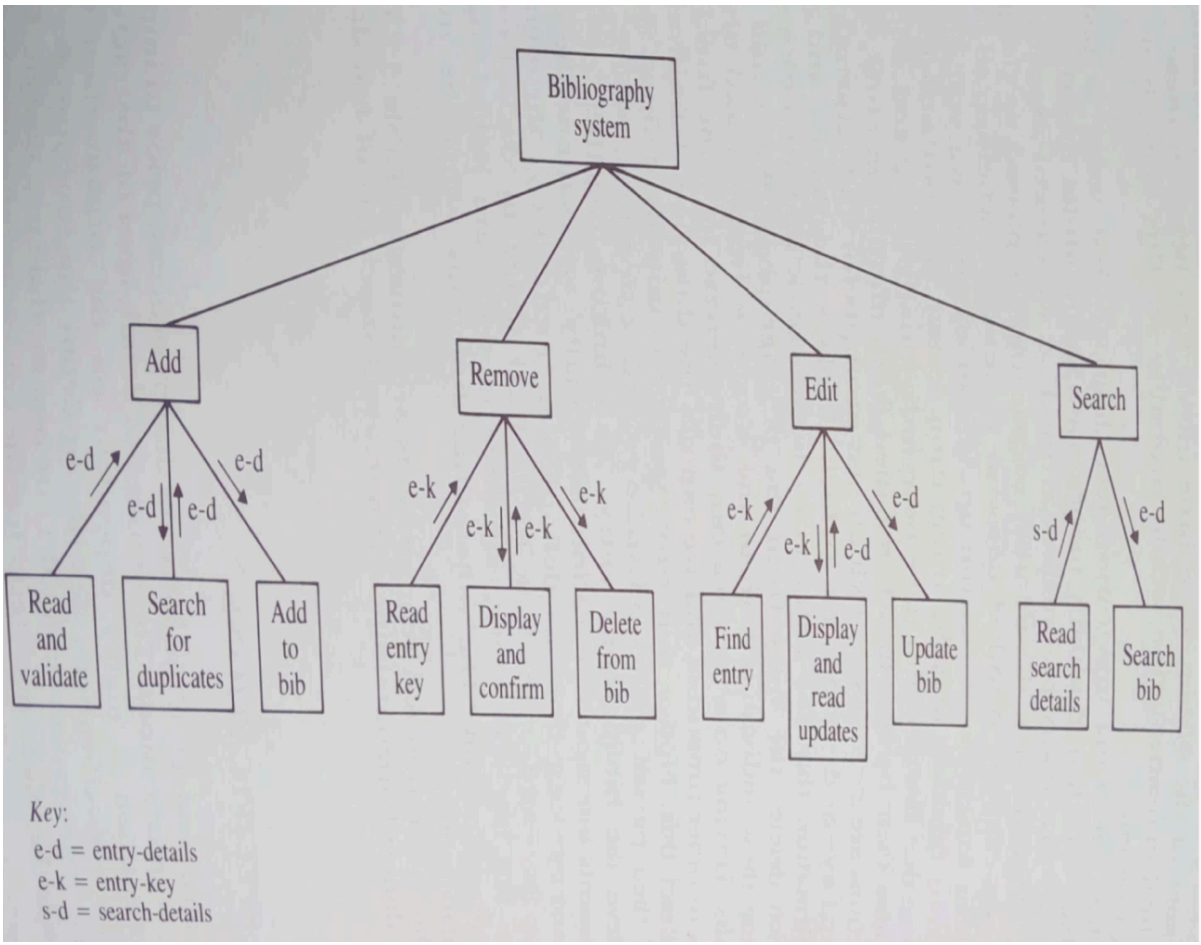

Figure 1: Architecture of Bibliography System

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Question 2 **20 Marks**

Consider the following raw description of a use case of an ATM. The use case is about establishing session at an ATM where an ATM card is inserted and a four digit PIN is entered.

Use case Description:

1. Wrong PIN entered, System displays 'Re-enter PIN'
2. Valid PIN entered but Bank is not online, System displays 'Try later' and ejects the card
3. Valid PIN entered, bank is online, account is closed, System displays 'Account closed, call your branch!' and ejects card
4. Valid PIN entered, Bank online, account is open, System opens Screen 2 and displays 'Select a service'
5. Invalid card, Bank online, System displays 'Card revoked' and retains the card
6. Invalid card, Bank not online, System displays 'Invalid card' and ejects the card

It might be noticed that Equivalence Class Partitioning (ECP) and Boundary Value Analysis (BVA) can help obtain input values for the test data (to test this use case).

Can decision tables also help design test cases for this use case? If yes, develop a decision table using the information available in the use case description. Clearly label the conditions, actions, and rules (rule number) etc. Complete all entries. Clearly state if any assumption is taken. The assumptions should be valid. If you have (somehow) developed a decision table then also provide a list of test cases that are derived from the developed decision table. How many test cases should be developed (give exact number). The test cases should have proper structure and include input values for all operational variables identified in the decision table. The test cases should also clearly express the expected output (for example ATM Message and Card Action). The test cases should be complete.

If decision tables cannot help design test cases for this use case, then give reason(s) why should we not use decision tables here. Once you have given the reason(s) only then provide test cases based on ECP and BVA. These test cases will be considered for grading only if the reason(s) for not using the decision table are considered valid, so try to give strong reason(s). What is the minimum number of test cases that should be developed (give exact number). The test cases should have proper structure and include input values for all operational variables involved in the scenario. The test cases should also clearly express the expected output (for example ATM Message and Card Action).The test cases should be complete.

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Question 3 **15 Marks**

Mature organizations use control charts to determine if their processes are under control or not. Control charts have been described in a Software Testing book as follows:

8.7.2 Control Charts

This statistical technique is an excellent tool for checking that processes are under statistical control. There are different types of control charts some for variable data and others for attribute data. The control chart is a graphical representation used to study how processes change over time. Data are plotted in a run chart with the average and control limits: upper control limit (UCL) and lower control limit (LCL) calculated from historical data.

By plotting current data on the graph one can conclude whether the process variation is consistent (in control seen by being within the band of UCL and LCL) or is unpredictable (out of control, affected by special causes of variation). When an out of control value is identified, it is marked on the chart and its cause investigated. The cause and its correction are documented for future reference.

For software statistical control the u-chart has been used to study the points that fall outside the control limits. The formulae for the u-chart are as follows:

$$UCL = \mu + 3\sqrt{stdev},$$
$$LCL = \mu - 3\sqrt{stdev},$$

where μ is the *weighted average defect rate from historical data* and stdev is the standard deviation.

A chart of component failure/KLOC is plotted against time in the Figure 8.3. As new points are collected they are plotted on the graph and the points above UCL are investigated. Points of excellence below LCL may also be identified to note the characteristics of the process of design/development. In drawing of these charts it is to be determined if the process under study is stable. If not the calculations based on the first set of points is only conditional and the limits and average have to be recalculated to be of use in the future.

When at least 20 sequential points from a period when the process is operating in control are obtained control limits are recalculated for optimal realization.

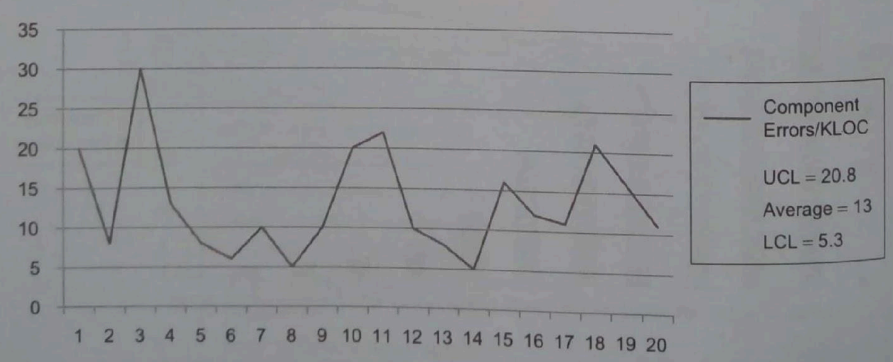

Figure 8.3: Control Chart Component Failures/KLOC with Time

Case Study:

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A company, My Development Company (MDC), develops web-based applications. MDC is applying statistical quality control during the testing process and tracking Mean Time BetweenFailures (MTBF) of the applications.

Different requirements regarding the applications that need to be met are as follows:

1. Search operation should complete within 3 seconds of the request 95% of the time
2. The application should work functionally correct 95% of the time
3. The application should run without failures 90% of the time

MDC tested the current version of an application for 10 days and corrected the faults during these 10 days. Every time the correction resulted in change in size of the application (measured in terms of Source Lines of Code or SLOC). The number of faults captured every day and the number of failures encountered every day are also recorded. The company has the following raw data collected during the testing phase:

Table 1: Raw data collected during performance testing

Time (Day)	SLOC	Faults	Failures
1	1705	62+(Last4DigitsOfYourRollNumber%10)	7-(Last4DigitsOfYourRollNumber%5)
2	1798	66+(Last4DigitsOfYourRollNumber%10)	8-(Last4DigitsOfYourRollNumber%5)
3	1776	96+(Last4DigitsOfYourRollNumber%10)	13-(Last4DigitsOfYourRollNumber%5)
4	1843	78+(Last4DigitsOfYourRollNumber%10)	17-(Last4DigitsOfYourRollNumber%5)
5	1925	66+(Last4DigitsOfYourRollNumber%10)	15-(Last4DigitsOfYourRollNumber%5)
6	1890	45+(Last4DigitsOfYourRollNumber%10)	11-(Last4DigitsOfYourRollNumber%5)
7	1875	50+(Last4DigitsOfYourRollNumber%10)	19-(Last4DigitsOfYourRollNumber%5)
8	2000	40+(Last4DigitsOfYourRollNumber%10)	14-(Last4DigitsOfYourRollNumber%5)
9	2080	55+(Last4DigitsOfYourRollNumber%10)	8-(Last4DigitsOfYourRollNumber%5)
10	2000	52+(Last4DigitsOfYourRollNumber%10)	10-(Last4DigitsOfYourRollNumber%5)

Redraw Table 1 before starting solving this question. After the 10 days period MDC needs to *determine if the application is stable in terms of failures or not*. They'll decide *whether they should go with further testing if the number of failures is not within certain limits*. They'd like to do it with the help of the control charts. Draw the control chart and draw conclusions that help MDC. Label the control chart clearly. The labeling should be complete. The conclusions should be useful and concrete, supported by the drawn and labeled control chart.

Hint: UCL, LCL, Average etc. might help draw conclusions.